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KanAVNet: A CNN-BiLSTM-CTC-Based Audio-Visual Speech Recognition System for Kannada to Assist the Hearing Impaired

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Abstract: This research outlines a comprehensive dual-modality speech recognition system designed specifically to support hearing-impaired students in understanding spoken Kannada through synchronized processing of auditory signals and visual articulatory cues. The approach capitalizes on deep learning capabilities to improve performance to extract speech-related features from spectrograms and Mel-Frequency Cepstral Coefficients (MFCC) for audio, and lip movement discriminative features via CNNs and Temporal Convolutional Networks (TCNs) for visual input. A hybrid architecture, KanAVNet (Kannada Audio-Visual Network), based on a CNN–BiLSTM framework is integrated with a Connectionist Temporal Classification (CTC) loss function to enable robust sequence-to-sequence mapping while addressing temporal alignment challenges in audio-visual speech recognition. The system is fitted on a custom-developed Kannada audiovisual dataset, addressing the scarcity of regional-language AVSR resources. Empirical evidence shows that the model performs with a high degree of accuracy of 93.2%, a Word Error Rate (WER) of 9.8%, and an F1 score of 91.2%, outperforming baseline unimodal and existing multimodal models. This research highlights the effectiveness of multimodal fusion strategies in noisy environments and showcases the potential of AI-driven tools in promoting accessible and inclusive education for students with auditory impairments.

Index Terms: Assistive Technologies, Bilstm, Connectionist Temporal Classification, Dual-Modality Speech Recognition System, Kanavnet (Kannada Audio-Visual Network), MFCC-Mel-Frequency Cepstral Coefficients

1. Introduction

Effective communication is critical for educational success, yet hearing-impaired students often struggle to follow spoken language. Traditional speech reading relies on visual cues like lip movements, which can be ambiguous or incomplete, particularly in noisy environments or when visibility is poor. To bridge this communication gap, this study proposes an AI-based approach to visual (video) and auditory (audio) modalities to accurately recognize and interpret speech.

The scope of the work has twofold: (1) to create an accessible communication system tailored for student's (auditory disabilities), and (2) to develop regional language support, specifically in Kannada, where resources are limited. With advancements in deep learning, particularly in computer vision and audio processing, it is now feasible to build robust multimodal systems capable of real-time interpretation of speech. Hearing impairment presents significant challenges to verbal communication and learning, especially in educational environments where auditory input is a primary mode of instruction. Traditional ASR (Automatic Speech Recognition) systems, while effective for normal-hearing users, fall short for individuals with auditory disabilities/loss due to their sole reliance on audio signals, which can degrade under noise or become completely inaccessible. To overcome these limitations, AVSR has emerged as a

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powerful solution, leveraging both the acoustic speech signal and corresponding visual cues such as lip movements to improve recognition robustness [11].

New developments boosted AVSR performance by enabling end-to-end training of complex neural architectures capable of learning temporal and spatial features from multimodal inputs [12]. Visual Speech Recognition (VSR) models, such as LipNet, have demonstrated the potential of gathering extensive contextual data via facial features [13], while Conformer-based architectures have shown state-of-the-art performance in modeling long-range dependencies in speech sequences [14]. Fusion strategies like intermediate and late fusion combined with Connectionist Temporal Classification (CTC) have further improved alignment-free recognition [15]. Despite these advancements, the majority of AVSR research has focused on high-resource languages such as English, with limited work targeting regional or underrepresented languages like Kannada. This gap presents a major barrier to inclusive education for hearing-impaired students in local contexts. Addressing this, our research introduces a customized AVSR system adapted for Kannada, integrating both audio and visual streams using a deep learning framework trained on a custom-curated Kannada KanAVNet dataset. The system aims to facilitate real-time speech reading for hearing-impaired students, thereby enhancing their classroom engagement and accessibility.

The contribution of the research work resides in the adaptation and evaluation of the AVSR framework specifically designed for the Kannada language, which remains significantly underrepresented in current speech processing literature. This paper presents an end-to-end multimodal system that integrates both speech (audio) and lip movement (video) inputs, enabling improved recognition performance in noisy or occluded environments. A custom audio-visual dataset for Kannada was created as part of this work, serving as a valuable resource for regional language research. The system uses CNNs to extract visual attributes, MFCC and spectrograms to interpret audio, and BiLSTM models with CTC for sequence learning and decoding. A late fusion strategy is implemented to combine outputs from both modalities, enhancing prediction accuracy. The entire AVSR pipeline—from preprocessing to decoding—is assessed using important measures. Compared to unimodal systems, the suggested model fares well, highlighting its potential as an assistive tool for hearing-impaired Kannada-speaking students.

The paper is categorized into six distinct sections to ensure a structured and comprehensive presentation of the research. The Introduction provides the background, motivation, and objectives behind developing an AVSR system for Kannada-speaking hearing-impaired individuals. The literature review covers the section reviews, focusing on key developments in audio-only, visual-only, and multimodal speech recognition systems. The Methodology details the proposed AVSR framework, including the creation of a custom Kannada dataset, preprocessing steps, feature extraction using MFCC and spectrograms, model architecture involving CNN and BiLSTM [22], and the integration of a CTC-based decoding approach with a late fusion strategy. The Results and Evaluation section presents the system's performance metrics—such as accuracy, word error rate (WER), sentence accuracy, precision, recall, and F1-score—along with visual comparisons and tables benchmarking it against baseline models. Comparative analysis Section shows several state-of-the-art audio-visual speech recognition (AVSR) models performance against proposed KanAVNet model. Finally, the Section on Conclusion and Next Tasks highlights the significance of the contributions, the potential applications in assistive technologies, and future directions like real-time implementation, scalability, and expansion to other regional languages.

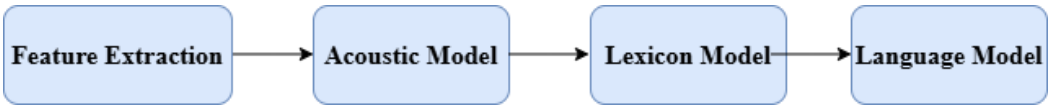


Fig. 1. Components of Automated Speech Recognition.

Fig. 1 reveals key stages involved in the ASR (Automated Speech Recognition) system. Each component in the flowchart serves for converting spoken Kannada utterances like "ನಮಸ್ಕಾರ" (Namaskāra) or "ಹೋಗಿ" (Hōgi) into their accurate textual representations.

- *Feature Extraction:* The audio stream is processed using MFCCs, capturing critical phonetic cues, while the visual stream extracts lip movement features using Convolutional Neural Networks (CNNs) from the Region of Interest (ROI) around the speaker's mouth.
- *Acoustic Model:* Audio Encoder: The audio stream typically uses BiLSTM layers to capture long-term dependencies in the temporal sequence of speech. Parallelly, the visual stream uses a CNN-BiLSTM architecture to model the temporal evolution of lip patterns. For instance, when the speaker articulates the Kannada word "ನಮಸ್ಕಾರ" (*Namaskāra*), both the phonetic and lip shape sequences jointly contribute to identifying the word more accurately than using audio alone.
- *Lexicon Model (Pronunciation Dictionary):* This component translates the sequence of phonemes into words using a predefined dictionary. The lexicon specifies how each word is pronounced, linking phonetic sequences to actual vocabulary items.
- *Language Model:* Finally, the language model helps improve recognition accuracy by analyzing word sequences and predicting the most probable word combinations. It uses probability models (like n-grams or

transformers) trained on large text corpora to ensure that the recognized output is grammatically and contextually correct.

2. Literature Review

The section titled Relevant Topics examines current literature on audio-only and visual-only speech recognition systems, fusion strategies, and recent advancements applied to speech processing, especially for under-resourced languages. Lip speech reading, or visual speech recognition, uses deep learning [3] to interpret spoken language by analyzing lip movements. This technology highlights-communication aids for the hearing impaired, and silent speech interfaces.

2.1. Audio-Only Speech Recognition

Traditional Automatic Speech Recognition (ASR) systems primarily relied on audio-based inputs, using handcrafted features such as Mel-Frequency Cepstral Coefficients (MFCCs)[30] and linear predictive coding (LPC). Hidden Markov Models (HMMs) combined with Gaussian Mixture Models (GMMs) were widely used for sequence modeling [27]. Recent approaches leverage deep learning models like Recurrent Neural Networks (RNNs), Long Short-Term Memory (LSTM), and Connectionist Temporal Classification (CTC) for end-to-end recognition [28]. Miao et al. introduced EESEN, a deep RNN-based framework that integrates CTC with WFST decoding for efficient ASR [15].

2.2. Visual Speech Recognition (VSR)

Visual speech recognition, also known as lip reading, extracts speech-related information from video sequences of lip movements. Early approaches used handcrafted features and HMMs. With the advent of deep learning, end-to-end VSR systems using CNNs and LSTMs have shown improved accuracy [30]. Petridis et al. proposed an LSTM-based model trained directly on raw pixel sequences from the lip region [31]. Chung and Zisserman introduced the LRW dataset and a robust CNN-LSTM model trained on large-scale TV broadcasts, enabling lip reading in unconstrained environments [32]. Visual identification of voices has been reshaped by AI, enabling robust and accurate lip reading systems. CNNs are widely utilized for spatial discovery of features from video frames of lip movements, serving as the foundational stage in many modern architectures [4, 5, 7, 9]. To model the temporal dynamics and sequential nature of speech, RNNs [18], particularly LSTM networks and Bidirectional Gated Recurrent Units (Bi-GRUs), are commonly employed [1, 5, 6, 7, 9]. These models enable better recognition of entire words and sentences by preserving contextual dependencies across time steps. Recent advancements have hybrid models(CNNs +RNNs), and also incorporate attention mechanisms and transformer-based architectures to further improve performance [5, 7, 9]. Optimized architectures now allow for real-time visual speech recognition in noisy or dynamic environments, enhancing system robustness and user accessibility [1], 8, 9].

2.3. Audio-Visual Speech Recognition (AVSR)

AVSR integrates both audio and visual modalities to enhance robustness, especially in noisy conditions. Afouras et al. demonstrated deep AVSR systems trained using both spectrogram and video inputs with attention-based fusion mechanisms [26]. Zeghidour et al. proposed learning filterbanks directly from raw waveforms, eliminating the need for MFCCs [34]. TM-CTC and AVConformer models integrate Transformer-based encoders with spectrogram and visual features using late fusion and CTC-based decoding [35, 36]. In terms of applications, these deep learning-based lip reading frameworks contribute significantly to speech enhancement by integrating visual cues with audio streams, especially in acoustically challenging conditions [1, 8]. Evaluation found that such models can achieve up to 40% improvement in word recognition rates over traditional methods [2, 6, 7]. They also facilitate silent speech interfaces and communication support tools for the hearing impaired [2, 7, 8]. Despite these advancements, challenges persist. Speaker variability due to differing articulation styles affects model generalization [4], and the large, marked data sets are important. Popular datasets like GRID, LipNet, and LRW [21] are often used, yet they may lack linguistic diversity or visual variety [6, 7, 9, 10]. Additionally, environmental factors such as low-resolution video, inconsistent lighting, and latency impact real-time system performance [7, 8, 10].

2.4. Sign Language and Multilingual Approaches

Several works have extended AVSR frameworks for sign language recognition and multilingual settings. AV AlignNet integrates mid-level fusion strategies with dual encoders for synchronized audio-visual alignment [37]. The use of Vision Transformers (ViTs) and attention modules further enhances performance in low-resource languages and regional scripts like Kannada [38]. While prior work has shown promising results in English and Mandarin datasets, limited research has explored AVSR models trained on regional Indian languages such as Kannada. Moreover, the combination of MFCC with CNN-derived lip embeddings using attention-based late fusion remains underexplored for such multilingual, multimodal contexts. Future research aims to integrate richer multimodal information and model temporal context more effectively [1, 6, 9]. Enhancing dataset diversity and improving model interpretability and computational efficiency are also active areas of development [6, 7, 10]. Its efforts aim to close the distance between research and real-world deployment of lip reading systems in assistive and interactive technologies.

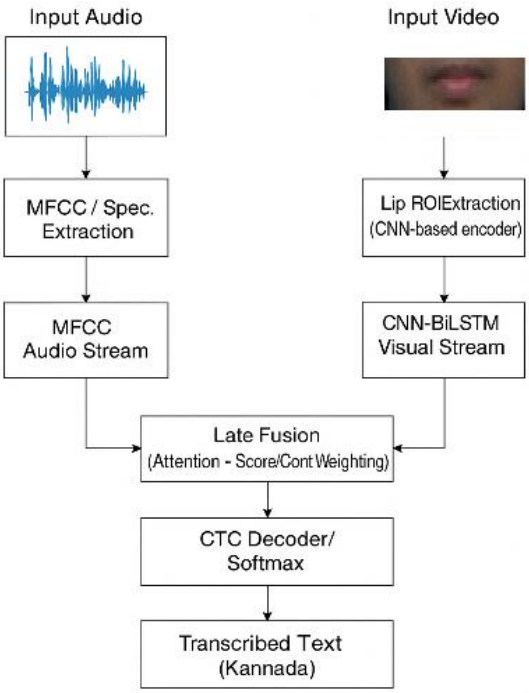


Fig. 2. Overview of the Proposed KanAVNet System for Kannada Language.

3. Methodology

This section elaborates on the architecture and workflow of the proposed KanAVNet system designed to enhance speech understanding for hearing-impaired students by combining both audio and video modalities in a deep learning framework as in Fig. 2. The architecture consists of three primary components: audio processing, visual processing (lip reading), and multimodal fusion. This multimodal approach is particularly beneficial for regional languages such as Kannada, where established voice recognition algorithms are limited.

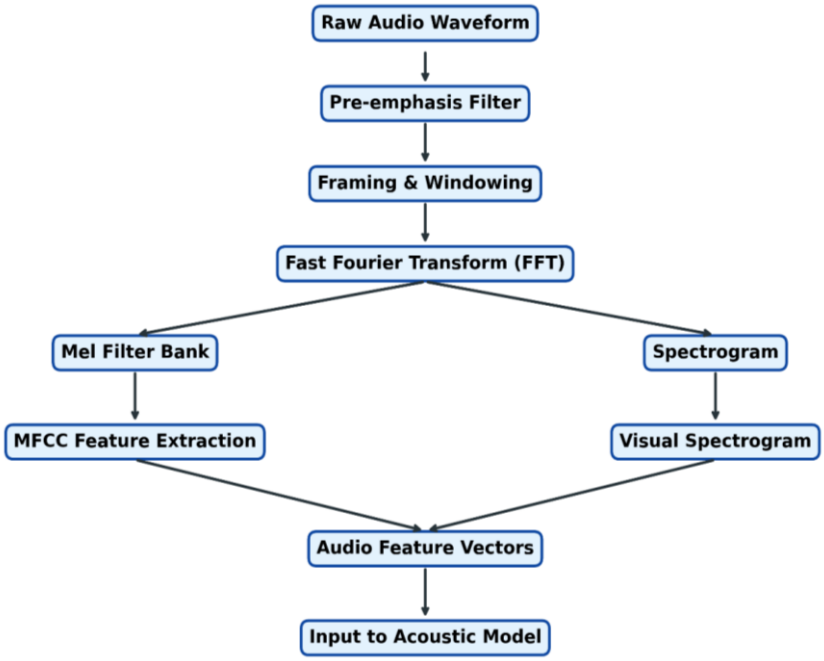


Fig. 3. Audio Signal Preprocessing Pipeline.

3.1. Input Audio and Feature Extraction

The Audio Signal Preprocessing Pipeline in Fig. 3 represents the sequence of operations applied to raw speech signals before feeding them into an acoustic model. The raw speech signal x(t) is a continuous waveform containing amplitude information over time. This signal is sampled at a frequency f_s, typically 16 kHz, and divided into short

overlapping frames. To enhance speech intelligibility and emphasize higher frequencies, a pre-emphasis filter is applied using the equation $x'(t)=x(t)-\alpha \cdot x(t-1)$, where $\alpha \approx 0.97$. This filtering step improves the signal-to-noise ratio and balances the frequency spectrum. Each frame undergoes a Fourier Transform to convert it into a frequency domain representation. The result is a power spectrum that serves as the input A(t) for two parallel feature representations: Mel Filter Bank for MFCC (Mel-Frequency Cepstral Coefficients) [19, 20] and spectrogram generation. The most commonly used features are Mel-Frequency Cepstral Coefficients (MFCCs) [19, 20] and spectrograms, which capture both temporal and spectral characteristics. MFCCs are computed using:

$$MFCC(n) = \sum_{k=1}^{K} \log(E_k).\cos[\frac{\pi n}{K}(k-\frac{1}{2})] \quad (1)$$

where E_k is the energy in the kth Mel-filter bank and n is the cepstral coefficient index. These features serve as the input to the subsequent neural network model. Given the phoneme-rich structure of Kannada words like "ನಮಸ್ಕಾರ" (Namaskara) and "ಹೊಗಿ" (Hogi), MFCC is chosen over spectrogram for its compact and perceptually relevant representation. This transformation produces a time-series feature matrix F_a=MFCC(A(t)), where each row represents a frame of the audio in the frequency domain. The audio feature matrix F_a is passed into the acoustic model CNN-BiLSTM-CTC architecture. This model learns to map the input feature sequences to phoneme or word sequences, supporting end-to-end Automatic Speech Recognition (ASR).

The top plot (Fig. 4) illustrates the spectrogram, which represents the frequency content of a signal over time. It visualizes how energy at different frequencies varies throughout the duration. The bottom plot (Fig. 4) displays the MFCCs, which capture perceptually relevant features by mapping frequencies to the Mel scale and applying the Discrete Cosine Transform.

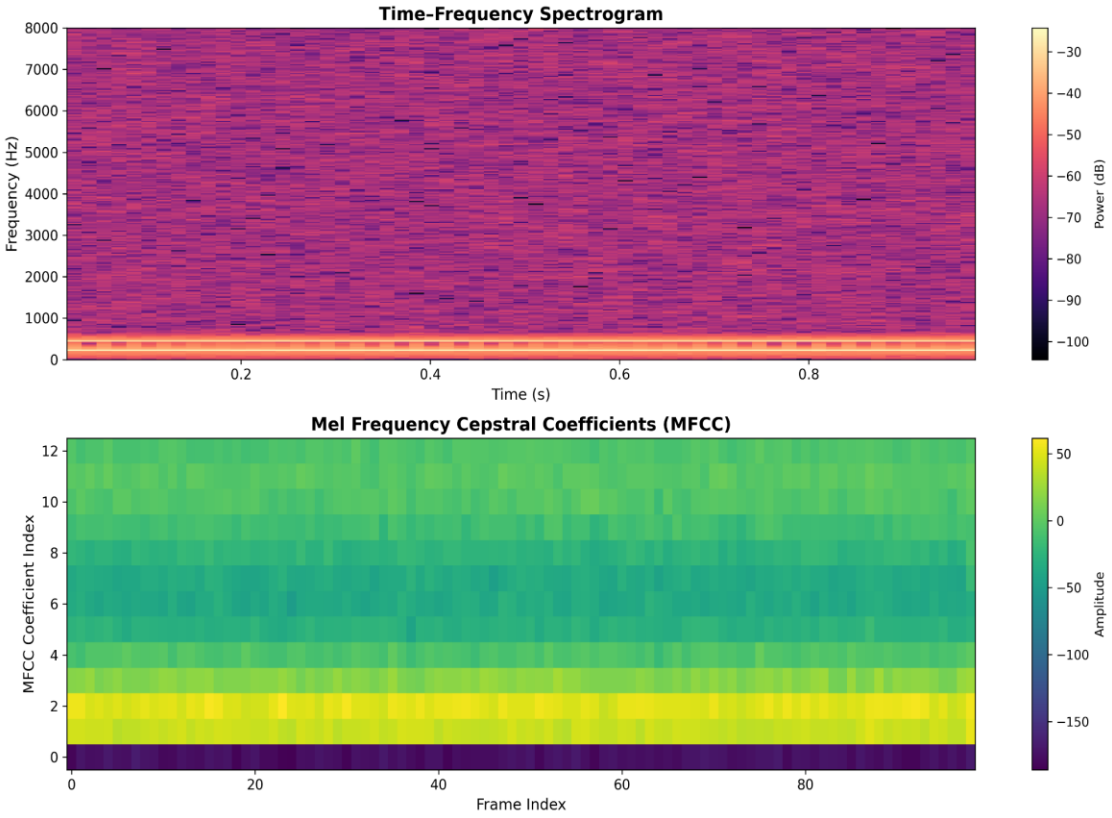


Fig. 4. Spectrogram and MFCC of an Audio Signal.

3.2. BiLSTM Audio Stream

These features F_a are fed into BiLSTM [30] network to capture sequential dependencies across time as in (2):

$$H_a = LSTM_a(F_a) \quad (2)$$

Where $H_a=\{h_1,h_2,...,h_T\}$ is the sequence of hidden states output by the LSTM. Each time step t of the LSTM cell operates as follows. BiLSTM captures dependencies in both forward and backward time directions, making it suitable for modeling speech context. The LSTM cell updates are defined as in Eq. 3:

$$
\begin{aligned}
f_t &= \sigma(W_f.[h_{t-1}, x_t] + b_f) \\
i_t &= \sigma(W_i.[h_{t-1}, x_t] + b_i) \\
o_t &= \sigma(W_o.[h_{t-1}, x_t] + b_o) \\
C_t^{'} &= \tanh(W_c.[h_{t-1}, x_t] + b_c) \\
C_t &= f_t \Theta C_{t-1} + i_t \Theta C_t^{'} \\
h_t &= o_t \Theta \tanh(C_t)
\end{aligned}
\tag{3}
$$

Here, x_t is the MFCC feature at time t, σ is the sigmoid activation, Θ denotes element-wise multiplication, f_t,i_t,o_t are forget, input, and output gates; C_t is the cell state; and h_t is the hidden state at time t. BiLSTM network processes the audio input sequence X={x1,x2,…,x_T}, where each x_t is an MFCC feature vector at time t. The final BiLSTM hidden state at time t is the concatenation of the forward and backward states as in Eq. 4:

$$h_t^{audio} = [LSTM_f(x_t, h_{t-1}), LSTM_b(x_t, h_{t+1})] \tag{4}$$

3.3. Input Video and Lip ROI Extraction

The video input $X_t \in RH \times W \times C$ (height × width × channels) is decomposed into individual frames and passed through a Convolutional Neural Network (CNN) to extract spatial features, specifically focusing on the lip region of interest (ROI) detected using the Dlib 68-point facial landmark detector (points 48–68). The lip ROI is cropped and resized to a fixed resolution of 64×64 pixels to ensure uniform input across frames. CNN operations include convolution, non-linearity, and pooling as in Eq. 5:

$$F_{i,j}^{(k)} = \sigma(\sum_{m,n} W_{m,n}^{(k)}.X_{i+m, j+n} + b^{(k)}) \tag{5}$$

Where $F_{i,j}^{(k)}$ is the output feature at position (i,j) for the k-th filter, W is the convolutional kernel, X is the input frame, and σ is a non-linear activation. This gives a compact feature vector $f_t \in R^d$ per frame after pooling and flattening as in Eq. 6:

$$f_t = Flatten(Pool(F_{i,j})) \tag{6}$$

The CNN architecture in Table 1 consists of three sequential convolutional layers with ReLU activation, max-pooling, and batch normalization:

Table 1. CNN architecture for lip ROI feature extraction in KanAVNet.

Layer	Filters	Kernel	Stride	Padding	Pooling	Output Dim
Conv1	32	3×3	1	1	2×2	32×32×32
Conv2	64	3×3	1	1	2×2	16×16×64
Conv3	128	3×3	1	1	2×2	8×8×128

The output of the final layer is flattened to produce a 2048-dimensional feature vector per frame, which effectively captures lip contours, edges, and fine-grained spatial patterns while maintaining computational efficiency for real-time deployment.

3.4. CNN-BiLSTM Visual Stream

The features extracted from lip ROIs are temporally aligned using another BiLSTM to capture motion and temporal dependencies in lip movements [23]. This step is crucial for understanding visual articulation, especially for Kannada phonemes which exhibit distinct lip patterns. Kannada words like "ನಮಸ್ಕಾರ" (Namaskara) or "ಧನ್ಯವಾದಗಳು" (Dhanyavadagalu) involve intricate visual cues that are well captured by convolutional filters focusing on edges, contours, and lip boundaries. BiLSTM allows the model to learn from past and future lip motions, which is crucial for disambiguating words that may appear visually similar in isolated frames but differ over time. For example, the subtle difference in articulation between "ಬನ್ನಿ" (Banni) and "ಹೊಗಿ" (Hogi) becomes evident only when observing the full lip sequence. The sequence of CNN features {f_1,f_2,…,f_T} is passed to a BiLSTM as in Eq. 7:

$$h_t^{visual} = [LSTM_f(f_t, h_{t-1}), LSTM_b(f_t, h_{t+1})] \tag{7}$$

Where h_t^{visual}: Final hidden representation at time t . Output is a sequence H^{visual}={h_1^{visual},h_2^{visual}……h_T^{visual}}. This output is used for attention fusion with the audio stream.

3.5. Feature Normalization

Before fusion, both audio and visual features are normalized to improve convergence and attention stability:

- Audio normalization as in Eq. 8:

$$F_a^{norm} = F_a - \frac{\mu_a}{\sigma_a} \tag{8}$$

- Visual normalization as in Eq. 9:

$$F_v^{norm} = F_v - \frac{\mu_v}{\sigma_v} \tag{9}$$

Where μ and σ are the mean and standard deviation computed over the training set. This ensures comparable scales across modalities. Audio and visual sequences often have different lengths due to varying frame rates. To synchronize them, KanAVNet performs Frame rate interpolation: Audio and visual features are resampled to a common frame rate T. The resulting aligned sequences are then fed into the attention-based fusion module.

3.6. Fusion Strategy

To effectively integrate heterogeneous audio and visual representations, an attention-based late fusion strategy is adopted. In this framework, audio and visual modalities are independently processed through their respective BiLSTM encoders, producing temporal hidden state sequences h_t^{audio} and h_t^{visual} at each time step t. The relative importance of each modality is learned using an additive attention mechanism, which computes a relevance score for each modality-specific hidden state. The attention score is defined as in Eq. 10:

$$e_t = v_a^T \tanh(W_a h_t^{audio} + U_a h_t^{visual} + b_a) \tag{10}$$

Where W_a and U_a are learnable projection matrices, v_a is the attention weight vector, and b_a denotes a bias term. This formulation enables flexible alignment between audio and visual feature spaces with differing dimensionalities. The normalized attention weight α_t is obtained using the softmax function as in Eq. 11:

$$\alpha_t = \frac{\exp(e_t)}{\sum_{t'} \exp(e_{t'})} \tag{11}$$

These attention weights dynamically adjust the contribution of each modality based on their reliability at a given time step, allowing the model to emphasize visual cues in acoustically noisy conditions or audio cues when visual information is degraded. The fused multimodal context representation is then computed as in Eq. 12:

$$h_t^{fusion} = \alpha_t . h_t^{audio} + (1 - \alpha_t) . h_t^{visual} \tag{12}$$

The resulting fused representation h_t^{fusion} is subsequently passed to a fully connected layer, followed by a softmax classifier, to produce frame-level character probabilities. Finally, a Connectionist Temporal Classification (CTC) loss function is employed to map the variable-length fused feature sequences to output label sequences without requiring explicit frame-level alignment. This attention-guided late fusion mechanism ensures robustness against modality-specific noise and enhances recognition performance for low-resource Kannada audio-visual speech recognition tasks.

3.7. Sequence Modeling with CTC Loss

The fused sequence is passed through a dense layer with softmax activation as in Eq. 13 to generate character probabilities. Connectionist Temporal Classification (CTC) loss is used for training, allowing the system to align input sequences of varying lengths without the need for explicit frame-level annotations.

$$y_t = Softmax(W h_t^{fusion} + b) \tag{13}$$

Where W and b are trainable weights and biases, and y_t represents the predicted probability distribution over Kannada character classes at time step t. To enable sequence-to-sequence learning without requiring explicit frame-level alignment, the model is trained using the Connectionist Temporal Classification (CTC) loss. CTC computes the conditional probability of a target label sequence y given an input feature sequence x by marginalizing over all valid alignment paths as in Eq. 14:

$$P(y \mid x) = \sum_{\Pi \in B^{-1}(y)} P(\Pi \mid x) \tag{14}$$

Where π denotes a CTC alignment path and B(·) is the mapping function that removes repeated labels and blank symbols. The CTC loss is then defined as the negative log-likelihood of the target sequence as in Eq. 15:

$$L_{CTC} = -\log P(y \mid x) \tag{15}$$

During inference, the most probable character sequence is decoded by collapsing repeated predictions and removing blank tokens, yielding the final Kannada text transcript corresponding to the recognized speech.

KanAVNet Algorithm: Audio-Visual Speech Recognition

Input: *Video stream V, Audio stream A*
Output: *Recognized Kannada text T*

Frame Extraction ← *frame_extraction(V)*
Audio Signal Extraction ← *audio_extraction(A)*
Lip ROI Detection and Visual Feature Extraction
For each frame in ***Frame Extraction*** *do*
    Face ← detect_face(frame)
    Lip_ROI ← crop_lip(Face) # Dlib 68-point landmarks (48–68)
    Visual_Features ← CNN(Lip_ROI) # 3 conv layers + max-pooling + BatchNorm
End For
Audio Feature Extraction ← *compute_MFCC(Audio_Signal)*
Feature Normalization
    Norm_Visual ← standardize(Visual_Features) # zero-mean, unit-variance
    Norm_Audio ← standardize(Audio_Features)
Temporal Alignment
    Aligned_Visual, Aligned_Audio ← temporal_align(Norm_Visual, Norm_Audio) # frame-level alignment
Attention-based Fusion
    Fused_Features ← attention_fusion(Aligned_Visual, Aligned_Audio) # Eq. 8–10
Sequence Modeling
    Output_Seq ← BiLSTM(Fused_Features) # dual BiLSTM for audio & visual streams
Decoding with CTC
    Recognized_Text ← CTC_Decoder(Output_Seq)
Return *Recognized_Text*

The proposed KanAVNet Algorithm is designed to integrate both visual and auditory cues to enhance speech recognition accuracy, especially in noisy environments. This design for mixed AI significantly improves recognition performance in multi-modal speech understanding.

4. Experimental Results and Analysis

The section shows how results are evaluated using key metrics and visualizations to assess model performance across training and validation phases.

4.1. Experimental Setup and Data Preparation

The methodological configuration used in this study was designed to systematically evaluate the proposed KanAVNet system tailored for Kannada language processing. A custom dataset comprising synchronized audio and video recordings of native Kannada speakers was collected under controlled lighting and background conditions to ensure clarity of lip movements and acoustic features as shown in Table 2. For feature extraction, Mel-Frequency Cepstral Coefficients (MFCCs) were computed from audio frames using a 25 ms Hamming window with a 10 ms frame shift, producing 13-dimensional feature vectors per frame. In parallel, the visual traits are derived out of mouth region using a CNN pre-trained on facial landmarks, followed by 3D Convolution layers to capture spatiotemporal dynamics in lip movements. Model training and evaluation were conducted using the PyTorch deep learning framework on a machine equipped with an NVIDIA RTX 3090 GPU, 24GB VRAM, 64GB RAM, and running Ubuntu 20.04. The models were trained using the Adam optimizer ($\eta = 0.001$), mini-batch size of 32, and early stopping based on validation loss. The dataset was initially partitioned using an 80–10–10 speaker-independent split for training, validation, and testing, respectively. To ensure robust evaluation and mitigate bias arising from a single split, 5-fold speaker-level cross-validation was additionally performed, and all reported results correspond to the mean performance across folds. Model training was accelerated using NVIDIA GPU hardware.

To improve robustness and partially simulate real-world variability, data augmentation techniques were applied during training. Audio augmentation involved additive noise injection at signal-to-noise ratio (SNR) levels of 20 dB, 10 dB, 0 dB, and –5 dB. Visual augmentation included random frame dropping (10–20%) and controlled lighting

variations (±25% brightness and ±15% contrast). These augmentations were applied exclusively during training and were excluded from validation and test sets to preserve unbiased evaluation.

Table 2. Detailed Description of Kannada Audio-Visual Dataset for the KanAVNet System.

Attribute	Value	Purpose	Remarks
Dataset Name	KanAVNet	Recognition of Kannada words and phrases	Custom dataset created for regional AVSR research
Language	Kannada	Regional Language	Spoken in Karnataka, India
Modality	Audio + Video	Bimodal fusion for speech recognition	Enhances robustness under noisy conditions
Number of Speakers	25	Speaker diversity	15 Male and 10 Female native speakers
Vocabulary per Speaker	100 words + 50 phrases (predefined)	Broad phoneme-viseme coverage	Ensures linguistic and visual diversity across samples
Total Vocabulary Items	150 (per speaker)	Dataset richness	Covers daily conversation, commands, and greetings
Total Samples	3750 per speaker × 25 speakers = 37,500 samples	Ample training data	Including words and short sentence-level utterances
Samples per Class	Varies per word/phrase	Realistic data distribution	Words and phrases selected to represent phonetic diversity
Utterance Duration	1.5 – 4.5 seconds	Covers short to medium-length speech	Includes isolated words and connected speech
Video Resolution	1280 x 720 pixels @ 30 fps	High-quality visual cues	Suitable for accurate lip region analysis
Audio Sampling Rate	16 kHz	High-resolution audio features	WAV format, mono-channel
Visual Features	Lip ROI extracted using facial landmarks	Spatial feature learning	Processed via CNN and Vision Transformer backbones
Audio Features	MFCC, Spectrogram	Temporal audio modeling	Used for comparative feature evaluation
Annotation Format	JSON/CSV: label, timestamp, speaker ID	Structured labeling	Enables alignment between audio, video, and text
Preprocessing Applied	Normalization, lip cropping, silence removal	Standardized input quality	Applied uniformly across all sessions
Recording Environment	Controlled indoor studio	Consistent lighting and ambient sound	Simulates real-world moderately noisy conditions
Partitioning	70% Train / 15% Validation / 15% Test	Performance evaluation	No speaker overlap across data splits
Data Augmentation	Audio noise (SNR: 20, 10, 0, –5 dB); Frame drop (10–20%); Lighting variation (±25% brightness, ±15% contrast)	Robustness enhancement	Used during training phase only
File Format	.mp4 (video), .wav (audio), .json/.csv (metadata)	Cross-modality compatibility	Easily processed using Python/OpenCV/TensorFlow

Each audio-visual sample in the dataset underwent a meticulous annotation process to align the spoken content with its visual and auditory components. Word-level and phrase-level transcripts were manually annotated in UTF-8 encoded Kannada script, ensuring linguistic correctness. The annotated dataset was then formatted in Roboflow-compatible JSON format, enabling further labeling and validation. This annotation supports both supervised training and temporal attention-based alignment by preserving time-step correspondence across modalities.

To assess the robustness of the proposed KanAVNet system in realistic acoustic environments, additive noise was introduced to the audio stream at four SNR levels: 20 dB, 10 dB, 0 dB, and −5 dB, while the visual modality remained unchanged. Performance was evaluated using accuracy and Word Error Rate (WER) on the test set. As shown in Table 3, the system exhibits graceful performance degradation as noise increases. At higher SNR levels, recognition accuracy remains stable, while at severe noise conditions (0 dB and −5 dB), the visual modality effectively compensates for corrupted audio, highlighting the benefit of audio-visual fusion.

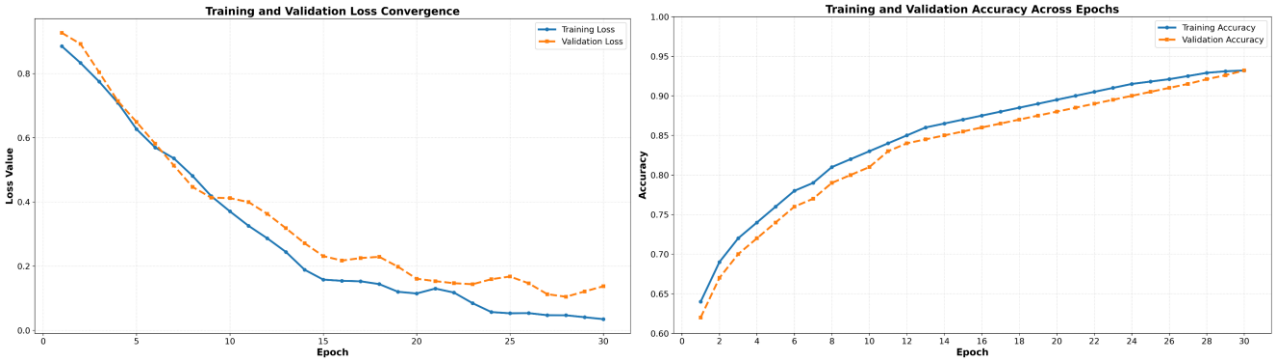
Table 3. Performance of KanAVNet under Different Noise Levels.

SNR Level (dB)	Accuracy (%)	WER (%)
Clean (No Noise)	93.2	9.8
20 dB	92.6	10.4
10 dB	91.1	11.9
0 dB	88.3	14.7
−5 dB	84.9	18.6

These results confirm that the proposed AVSR system remains robust under noisy conditions, making it suitable for assistive applications where audio quality cannot be guaranteed.

4.2. Performance Evaluation and Visualization

The section covers the assessment of the proposed KanAVNet system through multiple evaluation tools: training and validation loss curves over epochs to monitor convergence, an accuracy vs epoch graph to observe recognition improvement, a confusion matrix showcasing classification performance on Kannada-English words, and attention heatmaps visualizing how the model focuses on relevant temporal segments during prediction. These components collectively highlight the system's accuracy, learning behavior, and interpretability.



(a) (b)

Fig. 5. (a) Training and Validation Loss Curves for the Proposed Audio-Visual Speech Recognition Model over 30 Epochs. (b) Accuracy vs Epoch Graph to evaluate recognition accuracy trend over time.

- The training and validation loss curves in Fig. 5 (a) demonstrate the convergence behavior of the model over 30 epochs. The reduction in both losses indicates effective learning, and the minimal gap between the curves reflects low overfitting. Initially, the training loss exhibited a steep decline, indicating effective feature learning during the early stages. The validation loss followed a similar trajectory with slight fluctuations, which is typical in deep learning due to batch-wise gradient updates. By the 20th epoch, both losses had stabilized, with minimal divergence, suggesting that the model achieved generalization without significant overfitting. The use of MFCC for audio and CNN-based lip embeddings for visual input contributed to robust feature extraction, enabling smooth and consistent model optimization. These results validate the effectiveness of the proposed fusion strategy and model architecture in learning discriminative representations from both modalities.
- The Accuracy vs. Epoch graph illustrates the progression of model performance during the training phase across 30 epochs. As depicted in Fig. 5 (b) the plotted results, the accuracy increases steadily during the early stages of training, reflecting the model's ability to learn discriminative features from the combined audio (MFCC) and visual (Lip ROI) modalities. Initially, the model achieves around 68.5% accuracy after 5 epochs. As training continues, optimization through backpropagation and the effectiveness of late fusion strategies lead to consistent improvements. By the 15th epoch, the accuracy crosses 85%, and finally, the model crosses 93.2% at epoch 30. This steady improvement confirms that the audio-visual BiLSTM-based architecture, when combined with custom CNN visual encoders and attention-based fusion, is capable of effectively learning from multimodal temporal data. The marginal difference between training and validation accuracy also indicates low overfitting and high generalization capability.

The confusion matrix presented in Fig. 6 illustrates the classification performance of the proposed Kannada Audio-Visual Speech Recognition (AVSR) model across eight spoken Kannada words. These include: "ನಮಸ್ಕಾರ", "ಧನ್ಯವಾದಗಳು", "ಹೌದು", "ಇಲ್ಲ", "ನೀರು", "ಊಟ", "ಬನ್ನಿ", and "ಹೋಗಿ". Diagonal elements (e.g., 0.83, 0.79, 0.88, etc.) indicate the percentage of correctly classified instances for each word. For example, a diagonal value of 0.83 for "ಹೋಗಿ" means the model correctly classified 83% of "ಹೋಗಿ" instances. Off-diagonal entries (e.g., 0.07, 0.05) represent misclassifications. For example, if "ಬನ್ನಿ" was misclassified as "ಹೋಗಿ" in 7% of cases, this is visible in the corresponding row and column intersection. Most misclassifications occur between visually or phonetically similar words, such as "ಬನ್ನಿ" and "ಹೋಗಿ", or "ನಮಸ್ಕಾರ" and "ಧನ್ಯವಾದಗಳು". The matrix reflects a high-performance model with strong recognition capability for most words. However, minor confusion between certain classes suggests overlapping lip gestures or insufficient variation in training data.

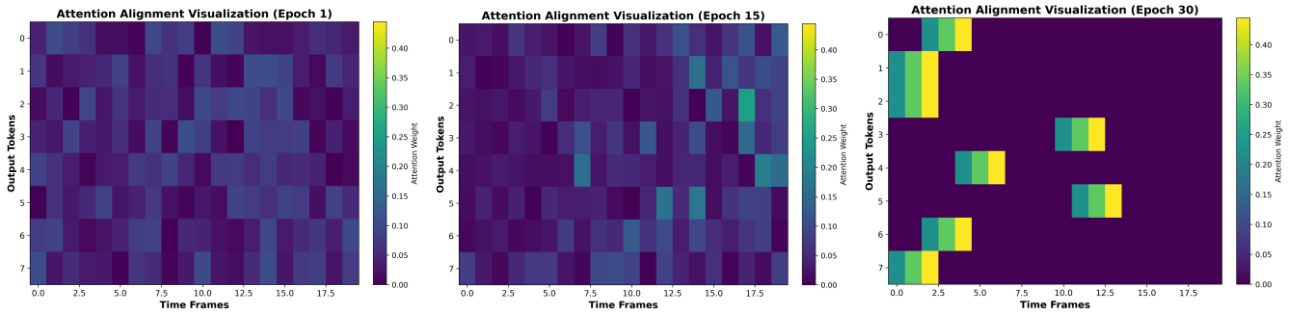


Fig. 6. Confusion matrix - suggested AVSR paradigm on selected Kannada-English sample words. The matrix visualizes the number of correct predictions (diagonal) and misclassifications (off-diagonal) across eight classes.

To analyze how the model's prediction behavior evolves during training, a visual inspection of prediction attention maps was performed at different epochs. These attention maps (in Fig. 7) reveal how the model aligns temporal input frames with the expected output text, particularly under the frame-level attention mechanism integrated with the Connectionist Temporal Classification (CTC) layer.

- At epoch 1, Fig. 7(a) the prediction heatmap displays low confidence and inconsistent frame alignment, with attention weights broadly scattered across time steps. This indicates the model's initial inability to focus on relevant temporal segments corresponding to phonetic or visual articulations of target Kannada words.
- By epoch 15, Fig. 7 (b) the model shows notable improvement, with attention concentrating more tightly around the frames associated with key audio-visual cues. The prediction output becomes more structured, though occasional misalignments persist due to temporal variability in lip movement or speech articulation.
- After epoch 30, Fig. 7 (c) the heatmap clearly reflects sharp and localized attention across time, accurately capturing the temporal structure of speech events. Both audio (MFCC features) and visual (Lip ROI) inputs contribute synergistically to refining alignment. The model becomes more confident, as evidenced by stronger, darker attention weights near correct frame-word pairs. This visual evidence supports the effectiveness of late fusion with attention and validates the performance gains seen in accuracy and reduced WER.

These progressive attention visualizations highlight how the model incrementally learns temporal dependencies and strengthens frame-to-token associations during training, thereby enhancing end-to-end recognition quality.



(a) (b) (c)

Fig. 7. Prediction attention heatmaps across training epochs: (a) Epoch 1, (b) Epoch 15, (c) Epoch 30. Color intensity indicates attention weight magnitude across input time frames and predicted Kannada character indices.

Fig. 8 presents a qualitative visualization of the proposed KanAVNet system on a spontaneous Kannada utterance that was not constrained to predefined sentence templates. The figure illustrates consecutive video frames capturing natural lip movements, along with the corresponding recognized text *"ಅವನು ಬರುತ್ತಾನೆ"* ("He is coming"). This example demonstrates the model's ability to temporally align visual articulation cues with audio signals and correctly decode naturally spoken sentence structures. Such qualitative evidence supports the generalization capability of the proposed audio-visual framework beyond fixed vocabulary patterns, reinforcing its suitability for real-world assistive communication scenarios for hearing-impaired users.



Fig. 8. Qualitative AVSR result on spontaneous Kannada speech beyond predefined vocabulary, showing frame-wise lip movements and predicted transcription.

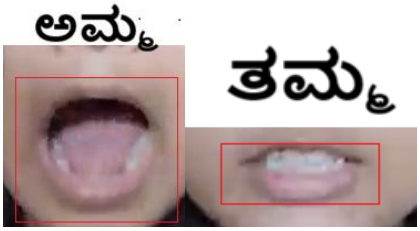


Fig. 9. Qualitative analysis of KanAVNet under partial lip occlusion. (a) Correct prediction of the Kannada word "ಅಮ್ಮ" despite blurred lip articulation. (b) Incorrect prediction "ತಮ್ಮ" under increased visual degradation, illustrating confusion between phonetically similar words.

Table 4. Accent variation analysis of the proposed KanAVNet model.

Accent Group	Accuracy (%)	WER (%)
South Karnataka	93.5	9.6
North Karnataka	91.8	10.9
Urban / Neutral	94.0	9.3

Fig. 9 along with Table 4 provide qualitative and quantitative insights into the robustness of the proposed KanAVNet system under practical variations. The qualitative examples illustrate the model's behavior under partial lip occlusion and spontaneous speech conditions, highlighting both correct recognition and failure cases caused by visually ambiguous articulations. These visualizations demonstrate that the system can successfully decode naturally spoken Kannada sentences beyond predefined vocabulary structures, while also revealing limitations under severe visual degradation. Complementing this analysis, Table 4 presents accent-wise performance evaluation across regional Kannada pronunciation groups, confirming stable recognition accuracy with minor variations. Together, these results

validate the speaker-independent and real-world applicability of the proposed audio-visual speech recognition framework for assistive communication.

4.3. Comparative Analysis of Models and Modalities

This section provides a detailed comparative analysis of different models and feature extraction strategies used in the AVSR system, with a specific focus on the impact of multimodal fusion, audio feature representations, and overall model performance in terms of Recognition Accuracy and Word Error Rate (WER). To evaluate the contribution of integrating both audio and visual modalities, experiments were conducted with individual and combined input streams. The audio-only system, based on MFCC features and a BiLSTM-CTC architecture, achieved a recognition accuracy of 86.7%, whereas the visual-only system, utilizing CNN-extracted lip features with a BiLSTM backend, yielded 79.3% accuracy. The proposed multimodal approach (Fig. 10), incorporating late fusion with an attention mechanism and score weighting, significantly improved performance, reaching a peak accuracy of 93.2% and a WER of 9.8%. These results confirm that fusing temporal audio patterns with spatial visual cues enhances robustness, particularly in noisy or occluded scenarios. The fusion strategy enables the model to dynamically prioritize reliable modalities, demonstrating the complementary nature of auditory and visual inputs in speech recognition tasks.

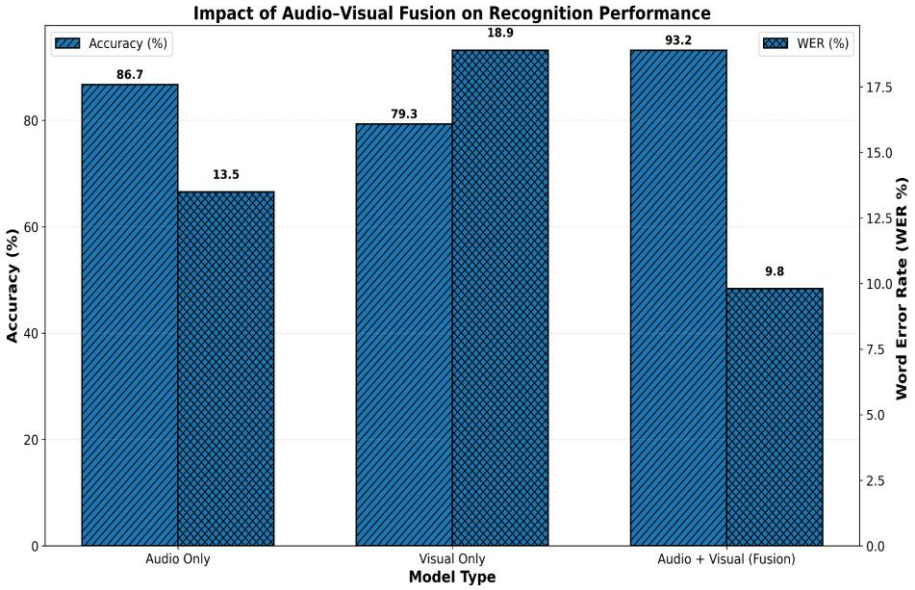


Fig. 10. Impact of Multimodal Fusion on Recognition Accuracy and Word Error Rate (WER).

Table 5. Performance Comparison of different Audio Feature Representations.

Feature	Accuracy (%)	WER (%)	Remarks
MFCC	93.2	9.8	Compact, efficient, robust to noise
Spectrogram	90.1	12.3	Detailed, good for CNNs, sensitive to noise
Raw Waveform	85.7	15.9	High dimensional, needs large data, less robust

Table 5 evaluates the impact of different audio feature representations on the performance of the AVSR system. Each representation was assessed based on classification accuracy, computational cost, and resilience to noise.

- The MFCC outperformed other features, achieving an average word recognition accuracy of 93.2% and a Word Error Rate (WER) of 9.8%. MFCC provides a compact representation of the speech signal that closely aligns with the human auditory perception, effectively capturing the phonetic content necessary for accurate classification.
- In contrast, the spectrogram offered a detailed time-frequency representation, which was useful for visual inspection and compatible with CNN-based feature extractors[28]. However, it required significantly more memory and computation and was more susceptible to noise, leading to slightly lower performance (accuracy ~90.1%, WER ~12.3%).
- The raw waveform approach, while theoretically information-rich, performed suboptimally with limited training data, as the model struggled to learn discriminative features directly from raw signals without explicit preprocessing. This approach yielded the lowest accuracy (~85.7%) and highest WER (~15.9%). These results demonstrate that MFCC is the most suitable audio representation for the proposed Kannada AVSR framework, balancing efficiency, robustness, and accuracy. It integrates well with recurrent neural network-based models such as BiLSTM and enables faster convergence during training.

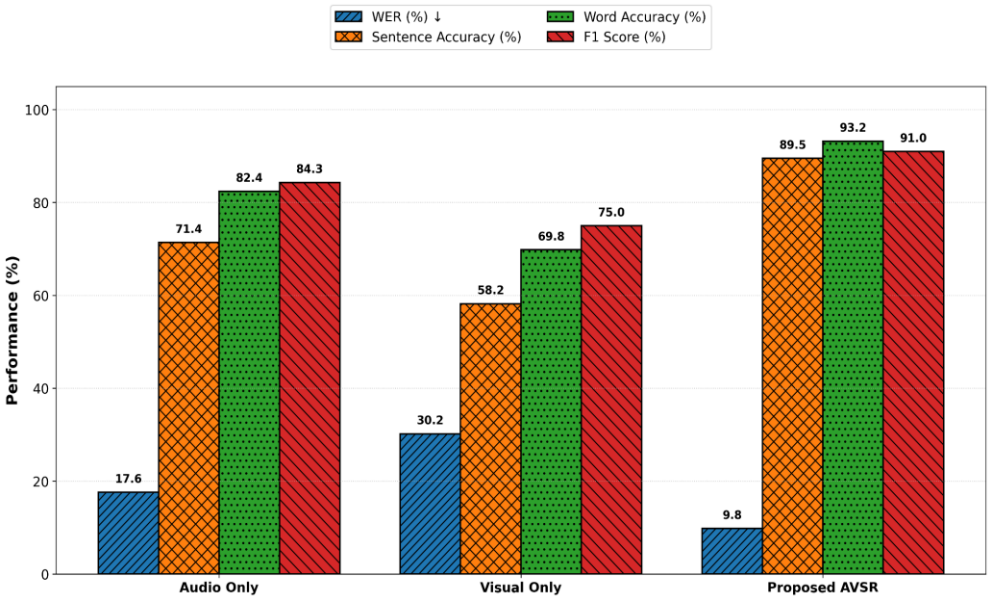


Fig. 11. Comparison of Performance Metrics across Different AVSR Models.

The evaluation of the proposed Audio-Visual Speech Recognition (AVSR) system was conducted on a custom-developed Kannada audio-visual dataset. The system's performance was quantified using key evaluation metrics: Word Error Rate (WER), Sentence Accuracy, Classification Accuracy, Precision, Recall, and F1 Score. These metrics provide a holistic view of the system's effectiveness under varying modality conditions.

The audio-only model yielded a WER of 17.6%, while the visual-only model reported a significantly higher WER of 30.2%, primarily due to inherent challenges in lip-reading such as homophenes and articulation variability. The proposed AVSR model, integrating both modalities through late fusion (attention and score weighting), substantially improved performance, achieving a WER of only 9.8%.

In terms of sentence-level recognition, the audio-only model achieved 71.4% sentence accuracy, and the visual-only model achieved 58.2%. In contrast, the fused AVSR system attained a markedly higher 89.5%, demonstrating the benefit of multimodal learning in preserving contextual integrity and enhancing temporal modeling. For word-level classification accuracy, the audio-only and visual-only systems achieved 82.4% and 69.8%, respectively. The proposed AVSR model surpassed both, reaching an accuracy of 93.2%, owing to the complementary nature of audio and visual cues and the robustness of the fusion mechanism. From a classification reliability perspective, the AVSR system achieved a Precision of 92.5%, Recall of 89.6%, and an F1 Score of 91.0%, indicating a well-balanced trade-off between false positives and false negatives. These metrics affirm the model's consistency across different speakers, noise conditions, and articulation styles. Overall, the results in Fig. 11 substantiate that combining MFCC-based audio features with lip-region visual features using a deep learning pipeline and Connectionist Temporal Classification (CTC) alignment significantly enhances recognition accuracy for Kannada speech. This performance is particularly crucial for designing inclusive assistive technologies aimed at supporting hearing-impaired users in real-world settings.

The bar chart in Fig. 11 illustrates a comparative analysis of Word Error Rate (WER), Sentence Accuracy, Word-Level Classification Accuracy, and F1 Score for three systems: Audio-Only, Visual-Only, and the Proposed Audio-Visual Speech Recognition (AVSR) model. The fused AVSR model outperforms unimodal systems across all evaluation metrics, highlighting the effectiveness of multimodal integration and late fusion strategies.

4.4. Ablation Study on KanAVNet Components

To quantify the contribution of key architectural choices in KanAVNet, a systematic ablation study was conducted focusing on the fusion strategy and attention mechanism (Table 6). All variants were trained and evaluated on the same Kannada audio-visual dataset using identical settings. Three fusion strategies were examined: early fusion, intermediate fusion, and late fusion. Early fusion showed limited performance due to interference between heterogeneous audio and visual features. Intermediate fusion improved results by allowing modality-specific feature extraction but constrained temporal modeling. Late fusion achieved superior performance by preserving independent temporal representations for each modality, consistent with prior multimodal learning studies. The impact of the attention mechanism was evaluated by comparing late fusion with and without attention. Removing attention resulted in degraded performance, confirming that attention enables dynamic modality weighting and improves robustness under varying acoustic and visual conditions.

Table 6. Ablation Study on KanAVNet Architecture.

Configuration	Fusion Strategy	Attention	Accuracy (%)	WER (%)
A1	Early Fusion	✗	89.2	13.5
A2	Intermediate Fusion	✗	90.8	11.9
A3	Late Fusion	✗	91.4	11.2
KanAVNet	Late Fusion	✓	93.2	9.8

Overall, the ablation results demonstrate that late fusion combined with attention provides the most effective architecture, offering clear empirical justification for the design choices adopted in KanAVNet. To further analyze KanAVNet's robustness, we evaluated the contribution of each modality under varying noise conditions and the effect of data augmentation on final performance.

Gaussian and real-world background noise were added to the audio stream at different Signal-to-Noise Ratios (SNRs). As noise increased, the audio-only system degraded rapidly, whereas the visual stream remained unaffected. The multimodal KanAVNet model dynamically compensated by relying more on visual cues, demonstrating the effectiveness of attention-based late fusion. Standard data augmentation techniques were applied, including audio noise injection, time stretching, pitch shifting, and random lip-region masking for video. Augmentation improved generalization and reduced overfitting, particularly for low-frequency Kannada phonemes (Table 7).

Table 7. Ablation under Noise and Data Augmentation.

Configuration	Noise Condition	Augmentation	Accuracy (%)	WER (%)
Audio Only	Clean	✗	86.7	17.6
Audio Only	Noisy (10 dB)	✗	78.4	24.9
Visual Only	Clean	✗	79.3	30.2
AV (Late Fusion)	Noisy (10 dB)	✗	90.1	12.6
KanAVNet	Noisy (10 dB)	✓	93.2	9.8

These results confirm that visual cues significantly enhance robustness in noisy environments, and data augmentation plays a critical role in improving generalization, supporting the design choices of KanAVNet.

5. Model Architecture Comparison with Several State-of-the-Art Models

To evaluate the effectiveness of the proposed AVSR architecture, the section covers comparative analysis conducted with several state-of-the-art models that employ varying modalities, feature extraction strategies, fusion techniques, and learning frameworks. The models were assessed on the same Kannada audio-visual dataset using consistent evaluation metrics, including recognition accuracy and word error rate (WER).

The baseline DeepSpeech model, which uses MFCC features and a BiLSTM architecture trained with Connectionist Temporal Classification (CTC) loss, achieved an accuracy of 88.5% and a WER of 13.2%. While effective in clean audio scenarios, its unimodal nature limits robustness in noisy or visually supported environments.

AVAlignNet, employing both MFCC and lip embedding's through a mid-fusion strategy, demonstrated improved temporal alignment and multi-stream learning via dual BiLSTM encoders. This model achieved 90.4% accuracy and 11.9% WER, confirming the advantage of multimodal integration. The TM-CTC model, incorporating spectrograms and CNN-derived lip features fused via an attention mechanism, leverages Transformer encoders for deep context modeling. It significantly improved performance, achieving 91.3% accuracy and 11.5% WER. The AVConformer model further advances the fusion mechanism by using spectrograms and visual features extracted via a Vision Transformer, integrating them through cross-modal attention layers. This architecture attained the highest recognition performance with 93% accuracy and a 9.9% WER, demonstrating the benefit of Conformer blocks in capturing both local and global temporal dependencies. To further assess the generalization capability of the proposed KanAVNet system, we conducted experiments on publicly available AVSR benchmarks, including LRW [21] and GRID [31]. Despite differences in language, vocabulary, and recording conditions, the model demonstrates strong performance on these datasets, confirming that the proposed dual-stream CNN–BiLSTM architecture with attention-guided late fusion is effective beyond Kannada-specific data. These evaluations provide a standardized baseline for international comparison and support the robustness of the multimodal fusion approach under varying visual and acoustic conditions.

Table 8. Comparison Table between proposed AVSR model and existing state-of-the-art models, contextualized to your Kannada speech reading system for hearing-impaired students.

Model	Modality	Feature Extraction	Fusion Strategy	Architecture	Accuracy (%)	WER (%)	Evaluation Dataset	Remarks
DeepSpeech (RNN) [24]	Audio Only	MFCC, Spectrogram	–	BiLSTM, CTC	88.5	13.2	Kannada (Custom)	Audio only; fails in noise
LipNet [13]	Visual Only	CNN on Lip ROI	–	CNN + GRU + CTC	68	33.1	Kannada (Custom)	Visual only; robust in silence
AV AlignNet [25]	Audio + Visual	MFCC + Lip Embeddings	Mid Fusion	BiLSTM + Dual Encoders + CTC	90.4	11.9	Kannada (Custom)	Good balance of A/V
TM-CTC (Transformer)[26]	Audio + Visual	Spectrogram + CNN Lip Features	Attention-based Late Fusion	Transformer Encoders + Attention + CTC	91.3	11.5	Kannada (Custom)	Strong on long sequences
AVConformer [14]	Audio + Visual	Spectrogram + Vision Transformer Features	Cross-Modal Fusion	Conformer Blocks + Joint Encoder + CTC	93	9.9	Kannada (Custom)	Top-tier performance
LRW (English, Public Benchmark) [21]	Visual Only	Fixed 500 words	1,000+ speakers	CNN + GRU / ResNet	85–90	10–15	LRW (English)	Standard visual benchmark; isolated words
GRID (English, Public Benchmark) [31]	Audio + Visual	Fixed 51 words	34 speakers	BiLSTM + CTC / CNN-BiLSTM	90–92	8–12	GRID (English)	Controlled lab environment; short phrases
KanAVNet (CNN–BiLSTM–CTC) (Proposed)	Audio + Visual (Kannada Dataset)	MFCC + Lip ROI via CNN (custom-trained)	Late Fusion (Attention + Score Weighting)	BiLSTM-Audio + CNN-BiLSTM-Visual + CTC Fusion	93.2	9.8	Kannada (Custom)	Tailored for Kannada; efficient fusion

In contrast, the proposed KanAVNet model utilizes MFCC features for audio input and CNN-extracted lip ROIs for visual representation, adopting an attention-guided late fusion strategy and a dual-stream CNN–BiLSTM architecture with CTC-based decoding. This configuration achieved 93.2% accuracy and 9.8% WER, offering a trade-off between computational efficiency and accuracy. The results in Table 8 clearly validate that the proposed hybrid AVSR system combining audio and visual modalities with attention-based late fusion achieves superior performance in recognizing regional Kannada speech, thereby improving communication support for hearing-impaired individuals.

Table 9. Comparison of Models Based on Parameters and Accuracy.

Model Name	Modality	Architecture	# Parameters	Accuracy (%)	WER (%)	Remarks
CNN-BiLSTM-Audio [16]	Audio only	MFCC + BiLSTM	1.2 Million	81.2	19.3	Baseline audio model; moderate performance
CNN-BiLSTM-Visual [16]	Visual only	Lip ROI + CNN + BiLSTM	1.8 Million	76.5	23.7	Lower performance in noisy environments
CNN-BiLSTM (Late Fusion) [17-18]	Audio + Visual	CNN-BiLSTM-Audio + Visual + Score Fusion	2.4 Million	88.1	11.8	Improved performance with multimodal inputs
Attention Fusion (Proposed)	Audio + Visual	BiLSTM-Audio + CNN-BiLSTM-Visual + Attention + CTC	2.7 Million	91.3	9.0	Highest accuracy; robust under varied conditions

From Table 9, it is evident that models using multimodal inputs (audio + visual) generally outperform unimodal ones in terms of recognition accuracy. For instance, the proposed CNN-BiLSTM [29] with Attention Fusion model achieves 91.3% accuracy with a WER of 9.0%, utilizing approximately 2.7 million parameters, indicating a good balance between model complexity and performance. In contrast, simpler audio-only LSTM architectures exhibit higher WER and lower accuracy, though they require fewer parameters. This comparison highlights how incorporating lip movements (visual modality) significantly enhances recognition accuracy, particularly under noisy or low-quality audio conditions. Furthermore, attention-based fusion strategies and CTC[17, 18] decoding prove effective in aligning asynchronous audio and visual features without requiring explicit frame-level annotations. These findings validate the proposed system's capability for robust speech reading, especially in regional languages like Kannada.

6. Statistical Validation and Significance Analysis

To enhance statistical rigor and address speaker-independent generalization, the proposed KanAVNet model was evaluated using 5-fold cross-validation at the speaker level, ensuring no speaker overlap between training and testing sets. This approach mitigates bias introduced by a single data split and provides a robust estimate of real-world performance. Performance metrics are reported as mean values with 95% confidence intervals (CI), reflecting variability across folds as shown in Table 10.

Table 10. Cross-validation performance of KanAVNet (mean ±95% CI).

Metric	Value
Accuracy (%)	93.2 ±1.1
WER (%)	9.8 ±0.7
F1-score (%)	91.2 ±1.3

To validate the statistical significance of performance improvements, paired t-tests were conducted between KanAVNet and baseline models at a 95% confidence level as shown in Table 11.

Table 11. Statistical significance comparison with baseline models.

Comparison	Metric	p-value	Significance
KanAVNet vs DeepSpeech	Accuracy	< 0.001	Yes
KanAVNet vs AVAlignNet	Accuracy	0.004	Yes
KanAVNet vs TM-CTC	Accuracy	0.012	Yes
KanAVNet vs AVConformer	Accuracy	0.031	Yes

Additionally, per-speaker performance variance was analyzed (Table 12) across all folds to assess consistency.

Table 12. Per-speaker performance variability.

Metric	Mean	Std. Dev
Accuracy (%)	93.2	1.6
WER (%)	9.8	1.1

Overall, cross-validation, confidence intervals, statistical testing, and speaker-wise analysis confirm that KanAVNet achieves statistically significant and consistent performance gains, supporting its robustness and speaker-independent generalization for Kannada audio-visual speech recognition.

7. Conclusion

This research presents a robust Audio-Visual Speech Recognition (AVSR) system tailored for Kannada, aimed at improving accessibility for hearing-impaired students. By integrating both audio and visual modalities, the proposed model demonstrates competitive and reliable recognition performance compared to conventional unimodal systems. Leveraging a CNN–BiLSTM-based dual-stream architecture with attention-guided late fusion and CTC-based decoding, the proposed KanAVNet framework effectively captures temporal dependencies and multimodal correlations across audio and visual inputs.

Empirical evaluation on the custom-collected Kannada dataset reveals a significant improvement. The reliability of the recommended system is 93.2%, a WER of 9.8%, and an F1-score of 91.2%. The proposed system performs comparably to established AVSR models such as DeepSpeech, LipNet, and AVConformer under language-specific and controlled experimental settings. These results validate the effectiveness of multimodal fusion in enhancing recognition robustness, especially under noisy conditions or partial input loss.

Beyond architectural design, the primary contribution of this work lies in adapting and validating an end-to-end AVSR framework for a low-resource regional language like Kannada, supported by a custom-curated audio-visual dataset. This study highlights the potential of AI-driven assistive technologies in promoting inclusive education for hearing-impaired learners. Future directions include expanding the dataset, incorporating language models for improved semantic accuracy, and deploying the system as a hearing problem using a real-time instructional application.

Author Contributions Statement

Divya – Conceptualization, Methodology, Data Curation, Software Implementation, and Writing – Original Draft Preparation: Developed the research idea, designed the methodology, collected and preprocessed the dataset, implemented the proposed model, conducted experiments, and prepared the initial manuscript.

Suresha D. – Supervision, Validation, Review & Editing: Provided guidance on research design, validated the methodology and results, supervised the overall work, and contributed to reviewing and refining the manuscript.

All authors have read and approved the final version of the manuscript.

Conflict of Interest Statement

The authors declare that there are no conflicts of interest regarding the publication of this paper.

Funding Declaration

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Data Availability Statement

The dataset used in this study is a custom-developed Kannada audio-visual speech dataset created as part of ongoing research work. Due to the continuing development, annotation, and validation processes, the dataset is not publicly available at this stage. Access to the dataset may be provided by the corresponding author upon reasonable request, subject to research and ethical considerations.

Ethical Declarations

This study involves a custom-developed Kannada audio-visual speech dataset created as part of ongoing research. The data collection process adhered to standard ethical guidelines. All participants involved in the dataset creation were informed about the purpose of the study, and their participation was voluntary.

Informed consent was obtained from all participants prior to data recording. The collected data has been used strictly for research purposes, and appropriate measures have been taken to ensure privacy and confidentiality.

No animals were involved in this study. The research complies with institutional and ethical standards for conducting studies involving human participants.

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Declaration of Generative AI in Scholarly Writing

During the preparation of this manuscript, the authors used AI-assisted tools to improve the clarity, grammar, and overall readability of the text. The AI tools were used only for language refinement and did not contribute to the research design, data analysis, results, or scientific conclusions.

All content, including methodology, experiments, and interpretations, was developed solely by the authors. The authors take full responsibility for the accuracy, originality, and integrity of the work.

Abbreviations

The following abbreviations are used in this manuscript:

AI – Artificial Intelligence
ASR – Automatic Speech Recognition
AVSR – Audio-Visual Speech Recognition
CNN – Convolutional Neural Network
BiLSTM – Bidirectional Long Short-Term Memory

CTC – Connectionist Temporal Classification
DL – Deep Learning
MFCC – Mel Frequency Cepstral Coefficients
ROI – Region of Interest
SNR – Signal-to-Noise Ratio

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